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Roll No. : .....

324455(24)

B. E. (Fourth Semester) Examination, April-May, 2016

(New Scheme)

(Elect. Engg. Branch)

ELECTRICAL POWER SYSTEMS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each unit is compulsory and each carry 2 marks. Part (b), (c) and (d) any two solve each carry 7 marks.

Unit-I

1. (a) Define skin effect. 2

(b) Discuss the concept of g.m.d. Use the concept to

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derive the expression for inductance of a single phase line.

(c) Derive the expression of external flux linkage and expression for the total inductance of single phase line.

(d) Define corona loss and explain the factor affecting corona.

Unit-II

2. (a) Define ferranti effect.

(b) What is sag and calculate the sag for a space of 200 m. If the ultimate tensile strength of conductor is 5758 kgf? The weight of conductor is 604 kgf/km. Allow a factor of safety of 2.

(c) Derive expression for ABCD Parameter for nominal T configuration of transmission line also draw phasor diagram.

(d) A 110 kV, 50 Hz three phase overhead line deriver's a load of 40 MW at 0·8 p.f. lag at receiving end has following constants.

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$A = D = 0{\cdot}98 \angle 3°$, $B = 100 \angle 25°$,

$C = 5 \times 10^{-4} \angle 80°$ Mho. Find sending end voltage, current and power.

Unit-III

3. (a) Define power factor and give it's advantage.

(b) Explain the method's of improving power factor.

(c) A single core cable has a core diameter of 13·5 mm and insulation thickness of 11·5 mm and material of insulation has specific resistance of 8×10^6 Mega ohm meter. Calculate the insulation resistance per km.

(d) A factory which has a maximum demand of 175 kW a power factor of 0.75 lagging is changed at RS 72 per kVA per annum. If the phase advancing compliment cost RS 120 per kVAR.

Find the most economical power factor at which the factory should operate. Interest and depreciation total 10% of the capacital investment on the phase advancing equipment.

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Unit-IV

(a) What do you understand by distribution system? 2

(b) Write short notes on the following : 7

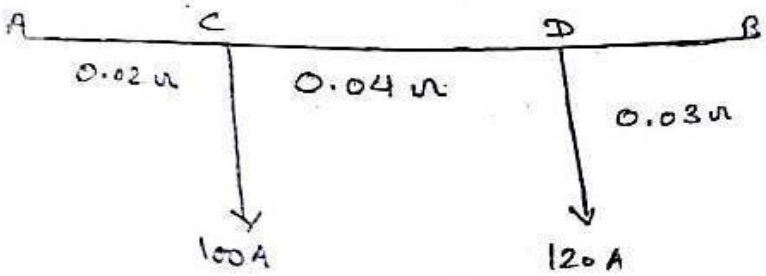
(i) Ring main system

(ii) 3-wire d.c. distribution

(iii) Radial system

(c) Derive the expression for maximum power transmitted by analytical method of synchronous phase modifier. 7

(d) A two wire distributor AB is fed at A and B at 250 V and 245 V respectively and is loaded as shown in fig. The resistance given are those for both conductor find the value of current in each section. 7



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Unit-V

5. (a) Why is Bewley's lattice diagram. 2

(b) Explain travelling wave phenomenon in open circuited and short circuited line. 7

(c) Show that voltage and current in a over headline travel's in form of wave. 7

(d) Derive an expression for the surge impedance of a transmission line. 7

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